Triblock Copolymers as Templates in Mesoporous Silica Formation: Structural Dependence on Polymer Chain Length and Synthesis Temperature

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Mesoporous silica can be formed with amphiphilic molecules acting as structure-directing templates in dilute systems. The templating properties of five related triblock copolymers, $(EO)_x-(PO)_y-(EO)_x$ (Pluronic L101, P103, P104, P105, and F108), have been investigated regarding the length of the EO blocks as well as the temperature of the synthesis. It was found that the number of EO units is essential in determining which silica mesophase is obtained. Lamellar structure is obtained with short EO chains (4 units); hexagonal structure, with medium length EO chains (17−37 units); and cubic structure, with long EO chains (132 units). The hydrophobicity of the $(EO)_x-(PO)_y-(EO)_x$ polymer is strongly affected by the temperature. This is to some extent reflected in the silica mesostructure formed; a multilamellar vesicle morphology can be obtained at elevated synthesis temperatures. Also, the temperature has an influence on particle size.

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Introduction

The interest in the field of mesoporous materials has expanded enormously since the first publications appeared in 1992−1993.[1−3] This is reflected in the number of papers related to mesoporous materials published per year.[4]

The synthesis of mesoporous materials relies on a supramolecular templating mechanism where an assembly of amphiphilic molecules, that is, surfactants, functions as a structure-directing template. Generally, the inorganic source is added to a dilute solution of the amphiphile, where the amphiphile normally would be organized in micelles.[5] The mesoporous material is formed in a precipitation reaction in which the micellar organization in most cases changes from its original form. For the most commonly obtained structure, MCM-41 with plane group *p*6*mm*, the micellar arrangement is rearranged with the addition of the inorganic source, resulting in a two-dimensional hexagonal silica structure.

Nonionic surfactants have been used as structure-directing agents in the synthesis of mesoporous silicas with disordered wormhole structures.[6,7] It was later realized that nonionic surfactants, among them triblock copolymers, had the ability to template ordered mesoporous materials, both siliceous and other inorganic phases.[8,9] The ordered silica phases obtained from syntheses with block copolymers have some advantages over the traditional M41S family (obtained in syntheses mediated by cationic surfactants). Mesoporous silicas formed using block copolymers are more stable than the M41S materials, probably due to the thicker silica walls obtained in the block copolymer mediated synthesis. The pore size is also affected by the method of formation. Block copolymers promote a significantly larger pore size. Zhao et al.[9] examined several polymers, including a range of triblock copolymers of the Pluronic family (vide infra) as templates in mesoporous silica synthesis. They obtained three ordered mesostructured silicas, lamellar with Pluronic L121, two-dimensional hexagonal (SBA-15) with for example Pluronic P123, and cubic (SBA-16) with for example Pluronic F127.

Block copolymers consisting of poly(propylene oxide) (PO) and poly(ethylene oxide) (EO) display amphiphilic properties in aqueous solutions. PO homopolymers phase separate from water at relatively low temperatures (approximately 5 °C), whereas EO homopolymers (that is, PEG, poly(ethylene glycol)) are well soluble in water. A block copolymer consisting of a central PO block (hydrophobic) surrounded by two EO blocks (hydrophilic) associates in aqueous solutions forming micelles consisting of a PO core and a corona of hydrated EO segments.[10] Further, some polymers self-assemble in water forming liquid crystalline phases.

A number of triblock copolymers are commercially available from BASF Corp. (USA) under the trade name Pluronics. By variation of the molecular weight and the EO/PO ratio, the amphiphilic character of the polymer can be modified. Thus, $(EO)_x-(PO)_y-(EO)_x$ amphiphiles can meet specific needs in various applications, for example, foaming, detergency, lubrication, and dispersion stabilization.[11]

The amphiphilic behavior of $(EO)_x-(PO)_y-(EO)_x$ polymers is temperature dependent. As the temperature is raised, the EO blocks become less soluble in water and hence the molecule becomes more hydrophobic. This

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(1) Beck, J. S.; Vartuli, J. C.; Roth, W. J.; Leonowicz, M. E.; Kresge, C. T.; Schmitt, K. D.; Chu, C. T.-W.; Olson, D. H.; Sheppard, E. W.; McCullen, S. B.; Higgins, J. B.; Schlenker, J. L. *J. Am. Chem. Soc.* **1992**, *114*, 10834−10843.
(2) Inagaki, S.; Fukushima, Y.; Kuroda, K. *J. Chem. Soc., Chem. Commun.* **1993**, 680−682.
(3) Kresge, C. T.; Leonowicz, M. E.; Roth, W. J.; Vartuli, J. C.; Beck, J. S. *Nature* **1992**, *359*, 710−712.
(4) Raimondi, M. E.; Seddon, J. M. *Liq. Cryst.* **1999**, *26*, 305−339.
(5) Cheng, C.-F.; Luan, Z.; Klinowski, J. *Langmuir* **1995**, *11*, 2815−2819.
(6) Bagshaw, S. A.; Pinnavaia, T. J. *Angew. Chem., Int. Ed. Engl.* **1996**, *35*, 1102.
(7) Bagshaw, S. A.; Prouzet, E.; Pinnavaia, T. *Science* **1995**, *269*, 1242−1244.
(8) Yang, P.; Zhao, D.; Margolese, D. I.; Chmelka, B. F.; Stucky, G. D. *Chem. Mater.* **1999**, *11*, 2813−2826.
(9) Zhao, D.; Huo, Q.; Feng, J.; Chmelka, B. F.; Stucky, G. D. *J. Am. Chem. Soc.* **1998**, *120*, 6024−6036.
(10) Zheng, Y.; Won, Y.-Y.; Bates, F. S.; Davis, H. T.; Scriven, L. E.; Talmon, Y. *J. Phys. Chem. B* **1999**, *103*, 10331−10334.
(11) Alexandridis, P. *Curr. Opin. Colloid Interface Sci.* **1996**, *1*, 490−501.

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anomalous behavior can be explained by considering the conformation of the oxyethylene group. At low temperature, a gauche conformation is prevailing, giving rise to a high dipole moment. As the temperature is raised, the group preferentially adopts the higher energy form which is the anti conformation with no dipole moment, thus giving rise to a more hydrophobic behavior.[12] This is apparent when the phase diagram of P104 in water is examined.[13] At room temperature, a 40 wt % water solution of P104 is cubic, but as the temperature is increased the structure adopts a hexagonal symmetry reflecting the more hydrophobic character of the polymer.

The type of liquid crystal structure formed in a binary water−polymer system has been shown to be dependent on the polymer cloud point[14] which, in turn, depends of the length of the EO blocks, provided the central PO block length is constant. The cloud point is the temperature where a liquid−liquid phase separation occurs. A low cloud point is obtained for a polymer with short EO chains, whereas long EO chains give rise to a more hydrophilic character and, hence, a cloud point at a higher temperature. The structures formed follow the trend from lamellar through hexagonal to cubic with an increase in cloud point. This can be explained by the packing parameter (v/a_0l_c). As the number of EO units increases, the headgroup area (a_0) increases and consequently the packing parameter is diminished. The value of the packing parameter determines the structure formed. The trend is the following: $<1/3$, spherical micelles (e.g., cubic structure); $1/3-1/2$, cylindrical micelles (two-dimensional hexagonal structure); $1/2-1$, flexible bilayers (vesicles); and ≈ 1, planar bilayers (lamellar). All these structures have been reported in mesoporous silica systems formed with block copolymers as structure-directing templates, with the exception of monolamellar vesicles. Multilamellar vesicles have been obtained in a few systems. Tanev and Pinnavaia reported a porous lamellar silica with vesicular particle morphology (MSU-V) obtained with neutral diamines;[15] mesostructured silica vesicles (MSU-G) were also produced with neutral gemini surfactants.[16] In systems with Pluronics serving as templates, vesicular morphologies have been obtained. Pluronic P123 can assist in forming these structures in an aerosol process,[17] by using dimethylformamide as a cosolvent[18] and also by raising the temperature of the reaction mixture (75 °C) and subsequently cooling to room temperature.[19] In all these cases, stable vesicles were formed.

In this work, we have performed a systematic study of five closely related Pluronic block copolymers. The polymers investigated all have the same or approximately the same number of PO blocks, but the number of EO blocks varies. Hence, we have studied how the structure-directing properties on mesoporous silica are affected by the change in EO block length. In addition, since the hydrophobicity of the polymers is dependent on temperature we have investigated how the structure-directing properties are affected by a temperature change in the synthesis step.

(12) Jönsson, B.; Lindman, B.; Holmberg, K.; Kronberg, B. *Surfactants and Polymers in Aqueous Solution*; John Wiley & Sons: Chichester, 1998.
(13) Wanka, G.; Hoffmann, H.; Ulbricht, W. *Macromolecules* **1994**, *27*, 4145−4159.
(14) Zhang, K.; Khan, A. *Macromolecules* **1995**, *28*, 3807−3812.
(15) Tanev, P. T.; Pinnavaia, T. J. *Science* **1996**, *271*, 1267−1269.
(16) Kim, S. S.; Zhang, W.; Pinnavaia, T. J. *Science* **1998**, *282*, 1302−1305.
(17) Lu, Y.; Fan, H.; Stump, A.; Ward, T. L.; Rieker, T.; Brinker, C. J. *Nature* **1999**, *398*, 223−226.
(18) Zhao, D.; Sun, J.; Li, Q.; Stucky, G. D. *Chem. Mater.* **2000**, *12*, 275−279.
(19) Corkery, R. W.; Fogden, A. Manuscript.

Table 1. Data on the Triblock Copolymers Used in This Work Obtained from www.basf.com/businesses/chemicals/performance

				cloud point/°C	
Pluronic	molecular weight	$(PO)_y$	$(EO)_x$	1 wt %	10 wt %
L101	3800	59	4	15	11
P103	4950	59	17	86	52
P104	5900	61	27	81	78
P105	6500	56	37	91	94
F108	14600	50	132	>100	>100

Table 2. Results of the Syntheses

Pluronic	synthesis temp/°C	structure	unit cell size/Å (as-synthesized)	unit cell size/Å (calcined)
L101	2	lamellar	111	
P103	55	hexagonal	118	101
P104	75	hexagonal	118	106
P105	85	hexagonal	128	115
F108	22	cubic	168	146

Experimental Section

Synthesis. The Pluronics were obtained as a gift from BASF Corp. (USA). For a more detailed description of each polymer, see Table 1. Tetraethyl orthosilicate (98%) was obtained from Acros Organics, and HCl was obtained from Merck.

The syntheses were all performed under similar conditions. The block copolymer was mixed with Millipore water and 4 M HCl in a Teflon bottle. This mixture was tempered to the required temperature. Tetraethyl orthosilicate (TEOS) was added under high stirring for about a minute after which the stirring rate was lowered. The time for precipitation to occur varied depending on the Pluronic used and on the temperature. After about 24 h of stirring, the bottle was placed in an oven at 80 °C for approximately 48 h. The sample was filtered and washed with water. Calcination was performed at 500 °C for 6 h in air.

The following amounts of each respective reactant were used: 1.9 g of polymer, 30−70 g of water, 5−45 g of 4M HCl, and 4 g of TEOS. The weight ratio of the surfactant to water solution was 2.5%. The total weight of solution (water and HCl) was kept constant at 75 g.

Characterization. The small-angle X-ray scattering (SAXS) experiments were done with a compact camera equipped with a linear position sensitive detector (MBraun, Graz) and a Seifert ID 3000 (3.5 kW) generator.

For transmission electron microscopy (TEM), the samples were dispersed in acetone and placed on holey carbon copper grids. Images were recorded with a Philips 120 Biotwin microscope operated at 120 kV equipped with a Gatan CCD camera.

Nitrogen adsorption/desorption isotherms at 77 K were recorded using a Micromeritics ASAP 2400 instrument. Pore size distributions were obtained from the N_2 adsorption branch, using the Barrett−Joyner−Halenda (BJH) method,[20] and surface areas were achieved by the Brunauer−Emmett−Teller (BET) method.[21]

Results

The five Pluronics investigated in this report were all able to act as templates for mesostructured silica. The common feature among these Pluronics is the length of the central PO block, consisting of approximately 57 units. The surrounding EO blocks vary in length: L101, 4; P103, 17; P104, 27; P105, 37; and F108, 132 EO units on each side of the PO block. The structure-directing properties of the polymers were found to be different (see Table 2). Hence, the length of the EO block is the essential factor in determining the structure formed. L101, which has very short EO chains, templated a lamellar structure (Figure 1c); P103, P104, and P105, with intermediate

(20) Barrett, E. P.; Joyner, L. G.; Halenda, P. P. *J. Am. Chem. Soc.* **1951**, *73*, 373−380.
(21) Brunauer, S.; Emmett, P. H.; Teller, E. *J. Am. Chem. Soc.* **1938**, *60*, 309−319.

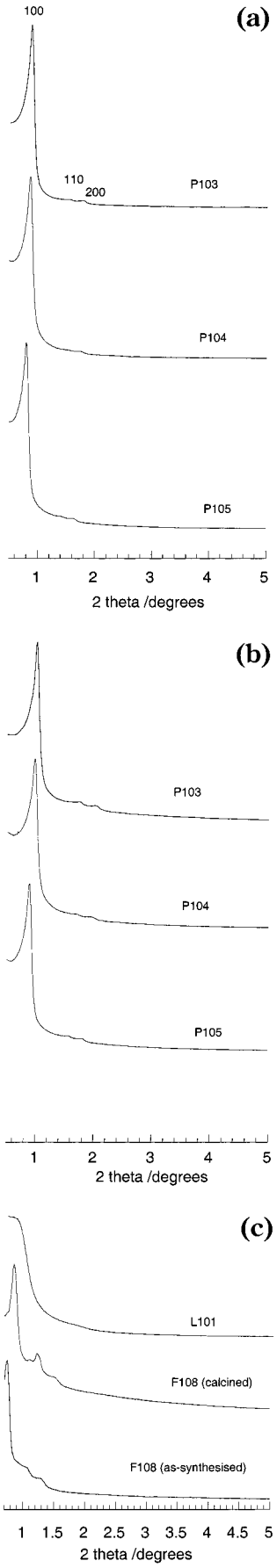


Figure 1. (a−c) SAXS patterns of P103−P105 (a) as-synthesized and (b) calcined and (c) F108 (as-synthesized and calcined) and L101 (as-synthesized).

Table 3. Results from Nitrogen Sorption Measurements of the Three Hexagonal-Forming Pluronics, P103, P104, and P105

Pluronic	synthesis temp/°C	surface area/ (m^2/g)	pore diameter (from adsorption pore volume plot)/Å	unit cell size (calcined)/ Å	wall thickness/ Å
P103	55	723	64	101	37
P104	75	712	65	106	41
P105	85	763	65	115	50

length of the EO blocks, preferentially templated a two-dimensional hexagonal structure (Figure 1a,b); and F108, with long EO chains, templated a cubic structure (Figure 1c). The polymer P103 showed an interesting behavior as the templating properties could be significantly modified by a change of temperature. This is due to the fact that the hydrophobicity of the polymer is temperature dependent. As the temperature is raised, the polymer gets more hydrophobic and the structure adopted will tend toward a less curved lamellar structure.

The nitrogen adsorption measurements shown in Table 3 indicate that the central PO block is the principle factor in determining the pore size. The pore size of the mesoporous silica templated by P103, P104, and P105 is essentially the same even though in these particular experiments the syntheses were performed at different temperatures. However, nitrogen sorption measurements of mesoporous silica obtained with P104 at three different temperatures, 55, 75, and 85 °C, all had the same average pore diameter (65 Å) indicating that the temperature does not affect the diameter although the pore size distribution was increased. The EO chain length on the other hand influences the wall thickness. Thicker pore walls are obtained with longer chains.

The temperature at which the cloud point occurs (see Table 1) was found to be the limiting temperature regarding the formation of ordered material. We have noticed that when the reaction mixture is heated to temperatures in the vicinity of the cloud point (which has been found to be slightly higher in acidic media than in pure water), the material formed is of bad quality and largely unaffected by the templating agent.

L101 ($(EO)_4(PO)_{59}(EO)_4$). The polymer with the shortest EO blocks gave rise to a lamellar structure. The quality of the structure improved with a temperature decrease from 12 to 2 °C. As the cloud point of this polymer is low (15 °C at 1 wt % water solution), it is not unexpected that a low temperature is needed for an ordered material to form. At the temperature of the cloud point, the polymer phase separates from the water solution. L101 contains only 4 EO groups on each side of the hydrophobic PO block. It is thus less soluble in water than the other Pluronics investigated here and has consequently a tendency to promote a lamellar structure.

Zhao et al.[9] synthesized hexagonal and lamellar silica with L121 at room temperature. The cloud point of this polymer occurs at 14 °C for a 1% solution and 10 °C for a 10% solution. This is a remarkable result as, according to the findings published here, L121 would not be able to template any mesoporous phase at room temperature, as this is high above the cloud point. Further, it seems unlikely that a hexagonal phase would be possible to synthesize at all with L121 as the number of EO blocks is low (5).

P103 ($(EO)_{17}(PO)_{59}(EO)_{17}$). This polymer has a very interesting behavior. It has moderately short EO chains (17 units) although long enough to form a hexagonal mesophase with silica at room temperature. Nevertheless, as the temperature is increased the hydrophobic character

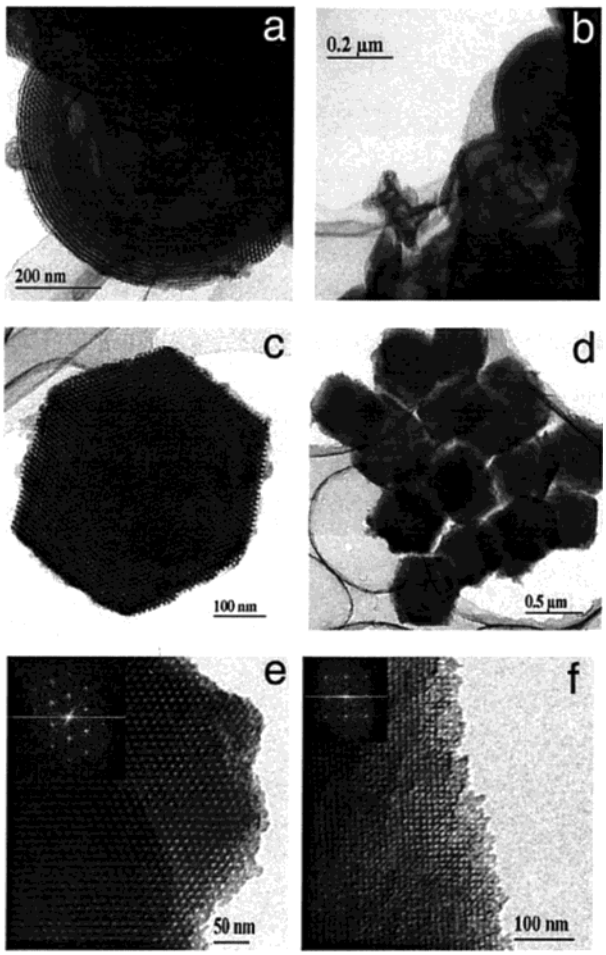


Figure 2. (a−f) TEM images of mesoporous silica formed with different block copolymers acting as templates: (a) and (b) show the mesoporous silica obtained with P103; (c) and (d), with P104; and (e) and (f), with F108.

becomes more pronounced which is reflected in the silica structures formed. At room temperature, the two-dimensional hexagonal SBA-15 is formed, but as the temperature is gradually increased the structure changes. At 45 °C, the particles are slightly curved; however, the hexagonal structure still prevails. At 55 °C, the sample contains two phases, the lion's share is the hexagonal structure but there is also a phase consisting of multilamellar vesicles (Figure 2 a,b). The multilamellar vesicles resemble the silica structures reported earlier for amphiphilic systems.[15−19] At 65 °C, the multilamellar vesicles are more numerous and well formed. At even higher temperatures, 75 °C, the material formed is less ordered. It consists of some lamellar regions, but the dominant part is material unaffected by the template. At 75 °C, the cloud point is reached and the solution is no longer a one-phase region resulting in the poorly formed silica.

It has also been observed that multilamellar vesicles can be obtained with P123 ($(EO)_{20}(PO)_{70}(EO)_{20}$).[19] Particularly well formed multilamellar vesicles are observed at 60°C. The concurrence in EO block length is probably the reason for the similar behavior of Pluronics P103 and P123.

A mixture of P103 and L101 (P103/L101 = 0.75/0.25 by weight) at 55 °C resulted in silica with a lamellar structure along with multilamellar vesicles. No hexagonal phase was detected. In this case, the L101, which is a lamellar former, directs the templating properties of the polymer mixture even more toward a less curved state than temperature alone can bring about.

P104 ($(EO)_{27}(PO)_{61}(EO)_{27}$). This polymer also formed the two-dimensional hexagonal structure of SBA-15 at room temperature. A temperature increase did not give rise to such a strong effect as was observed for P103. However, the quality of the hexagonal structure was improved. At 65 °C, the particles start to curve slightly. This behavior gets more pronounced at 75 and 85 °C. Nevertheless, P104 is not able to template vesicles or a lamellar structure because of the length of the EO blocks, at least not at the reactant concentrations used here. Still, another interesting effect of temperature was observed. The size of the hexagonal particles changed with temperature, and at 55 °C, which appears to be the optimum temperature, a very homogeneous particle size (about 0.5 μm in diameter) was discovered (Figure 2c,d). As the temperature was increased, the aspect ratio of the particles changed. They were still 0.5 μm long but somewhat thinner. This resulted in slightly curved particles, although the hexagonal structure still remained and no lamellar phase was detected even at 85 °C where the sample contained significant amounts of untemplated material as the cloud point had been reached.

When the stirring rate of the reaction mixture was examined for the 55 °C experiment, it was concluded that the size variation of the particles formed was enhanced by a lower stirring speed. It was also concluded that the actual size of the particles changed. Slow stirring gave particles in the 0.4 μm × 0.07 μm size regime. However, if the stirring speed was high for the first minute after TEOS addition all particles were of the same size regardless of the continued stirring speed. This implies that the particle aggregates are formed within the first minute of synthesis. The actual polymerization of the silica has not started, as the solution is still transparent. The precipitation begins after approximately 15 min.

P105 ($(EO)_{37}(PO)_{56}(EO)_{37}$). The polymer with 37 EO groups on each side of the central block behaved in many ways similar to P104. The structure obtained was two-dimensional hexagonal, and a gradual increase in temperature up to 95 °C never induced another structure. The particle size became more homogeneous and also smaller when the temperature was increased. However, the particle size was not as controllable as in the case of P104. The most homogeneous particle size was obtained at a temperature of 85 °C. All experiments were done below the cloud point, which occurs above 90 °C for P105.

Zhao et al.[9] observed an increase in pore volume with an increase of oven temperature. Also, the wall thickness diminished, whereas there was no clear trend for the $d(100)$ spacing. Figure 3 shows the unit cell parameter (a) as a function of synthesis temperature for the three hexagonal formers (P103, P104, and P105) that were used in this work. The temperatures are between room temperature (around 22 °C) and 95 °C. All values are not included due to the restricted temperature range where synthesis can be performed. Also, only the unit cell parameters that could be accurately calculated from diffractograms of sufficient quality are included. In general, the values of a increase with temperature for all three polymers.

F108 ($(EO)_{132}(PO)_{50}(EO)_{132}$). The number of EO groups on Pluronic F108 is considerably larger than on the other polymers in this study. It could thus be expected to act as a cubic former. This was confirmed by experiments (Figure 2e,f). Pluronic F127 ($(EO)_{106}(PO)_{70}(EO)_{106}$) templates for the cubic SBA-16 structure,[9] the structure of which recently has been determined by electron crystallogra-

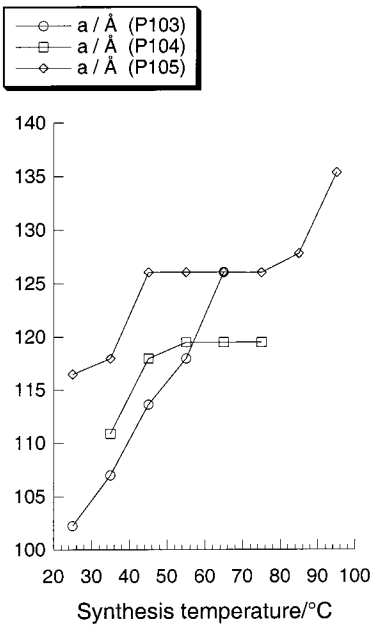


Figure 3. The dependence of unit cell parameter (*a*) on synthesis temperature for the three Pluronics (P103, P104, and P105) that form a hexagonal mesophase with silica.

phy.[22] The space group was confirmed to be *Im*3*m*. The lattice parameter for the as-synthesized sample of SBA-16 when templated by F127 is 124 Å.[9] Here, the cubic structure obtained has a larger lattice parameter, 168 Å for the as-synthesized structure. This is expected, as the total length of the polymer chain is longer in the case of F108. The X-ray diffraction pattern coincides with space group *Im*3*m* (Figure 1c).

The amount of acid needed for mesoporous silica formation was significantly smaller for F108 than for the other polymers: 5 g of 4 M HCl instead of 30 g which was used in the syntheses with P103, P104, and P105. L101 required the largest amount of HCl (45 g). This possibly reflects the attachments of protons to the outer EO groups of each polymer, as was postulated.[9] Since the weight fraction of the polymers to solution was kept constant, the number of moles of the polymers varied. Thus, the L101 syntheses contained the largest amount of polymer and the F108 the least.

Attempts were made to increase the temperature of the reactions of F108 though we obtained only an ordered mesoporous material at room temperature.

Discussion and Conclusions

The tendency for $(EO)_x-(PO)_y-(EO)_x$ block copolymers to adopt different structures depending on temperature as well as composition (EO/PO ratio) is well documented for the pure water polymer case.[13,14,23,24] We can now conclude that the same tendency is found for the systems where mesoporous silica is formed. Polymers with short EO chains form lamellar structures; with medium length chains, two-dimensional hexagonal structures are preferentially formed; and finally, long EO chains favor a cubic structure.

This work clearly demonstrates that the length of the EO chains is essential for establishing the mesoporous structure. At room temperature (or below in the case of L101), we have shown that only the short (4 EO) chain polymer, L101, gives rise to a lamellar structure, two-dimensional hexagonal (*p*6*mm*) is formed with the medium (17−37 EO) chain polymers, P103, P104, and P105, and the long (132 EO) chain polymer, F108, favors a cubic structure.

These findings corroborate those reported by Zhao et al.[9] In that report, it was concluded that the ratio of the number of units of the two monomers (EO/PO) determines the structure formed. The polymers with long EO chains (F127, F88, and F68) all formed cubic structures. The medium chain polymers (L64, P65, P123, and P103) formed hexagonal structures. The only short-chain polymer investigated (L121) formed lamellar structures. However, they also reported that L121 could form a hexagonal structure. Whether it is the EO/PO ratio or only the actual length of the EO chains that determines the structure formed is not possible to determine from the data mentioned, though it is confirmed that the EO chain length is essential for the structure formed.

We have also shown that the synthesis temperature is important for the structure-directing properties of the polymers. As the temperature is increased, the polymer becomes more hydrophobic which, at least in the case of P103, results in a structure preferentially formed in more hydrophobic systems, that is, multilameller vesicles. The packing parameter of vesicles ($^1/_2$−1) lies between that of cylindrical micelles (two-dimensional hexagonal) and that of a lamellar structure.

Further, the temperature had an effect on particle size as well as unit cell size. The unit cell parameter of the hexagonal silica structures increased with increasing synthesis temperature. For the syntheses performed with P104 and P105, the particle size along with the particle size distribution decreased with increasing synthesis temperature. This effect was most pronounced for the material formed with P104. A very homogeneous size distribution was obtained at 55 °C. For the synthesis with P105, the temperature at which a rather homogeneous size distribution was obtained (85 °C) was too close to the cloud point (90 °C). To achieve an even more homogeneous distribution, a higher temperature is needed which is not feasible due to the value of the cloud point. For the P103-mediated material, an increased temperature resulted in another morphology; thus, the particle size could not be controlled. For the structures formed with L101 and F108, no effect on particle size was obtained.

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(22) Sakamoto, Y.; Kaneda, M.; Terasaki, O.; Zhao, D.; Kim, J. M.; Stucky, G.; Shin, H. J.; Ryoo, R. *Nature* **2000**, *48*, 449−453.

(23) Alexandridis, P.; Zhou, D.; Khan, A. *Langmuir* **1996**, *12*, 2690−2700.

(24) Alexandridis, P. *Curr. Opin. Colloid Interface Sci.* **1997**, *2*, 478−489.